

Clustering Performance Analysis



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Clustering Performance Analysis

K-Clusters : $\mathcal{S}_1, \dots, \mathcal{S}_r, \dots, \mathcal{S}_K$

Data Instance $\mathbf{x}_i$ in Cluster $\mathcal{S}_r$: $\mathbf{x}_i^{(r)}$

Number of Data Instances in Cluster $\mathcal{S}_r$: n_r

Mean Vector ($\boldsymbol{\mu}_r$) and Covariance Matrix ($\mathbf{C}_r$) for Cluster $\mathcal{S}_r$ can only be computed for Numerical Vector Data Instances $\mathbf{x}_i^{(r)} \in \mathbb{R}^d$

Intra-Cluster Variation: How Compact are the Clusters?

Inter-Cluster Distance: How Separated are the Clusters?

Intra-Cluster Variation

Cluster $\mathcal{S}_r$ is *More Compact Compared* to Cluster $\mathcal{S}_j$

$$|\mathcal{C}_r| < |\mathcal{C}_j|$$

Cluster $\mathcal{S}_r$ has *Less Intra-Cluster Variation Compared* to Cluster $\mathcal{S}_j$

Worst Case: Maximum Intra-Cluster Variation

$$\max_{r=1,\dots,K} |\mathcal{C}_r|$$

Inter-Cluster Distance

Distance between Cluster $\mathbf{S}_r$ and Cluster $\mathbf{S}_j$

$$d(\mathbf{S}_r, \mathbf{S}_j) = \frac{1}{2} (\boldsymbol{\mu}_r - \boldsymbol{\mu}_j)^T (C_r^{-1} + C_j^{-1}) (\boldsymbol{\mu}_r - \boldsymbol{\mu}_j)$$

There are $\frac{K(K-1)}{2}$ such Distances for K clusters

Worst Case: Minimum Intra-Cluster Distance

$$\min_{[j,r \in \{1, \dots, K\}] \wedge [j \neq r]} d(\mathbf{S}_r, \mathbf{S}_j)$$

Using Mean Vector and Covariance Matrix

$$J(k) = \frac{\min_{[j,r \in \{1,\dots,K\}] \wedge [j \neq r]} d(\mathbf{S}_r, \mathbf{S}_j)}{\max_{r=1,\dots,K} |C_r|}$$

Higher Value Signifies Formation of Compact Clusters Far from Others

Data Instances may or may not be in Numerical Vector Space but their Distances are known

Silhouette Measure

The Perspective from Data
Instance $\mathbf{x}_i^{(r)}$ of Cluster $\mathcal{S}_r$

Silhouette Measure: Intra-Cluster Variation

$a_i^{(r)}$ is the Average of Distances between $\mathbf{x}_i^{(r)}$ and other Data Instances in Cluster $\mathbf{S}_r$

$$a_i^{(r)} = \frac{1}{n_r - 1} \sum_{\substack{j=1 \\ j \neq i}}^{n_r} d \left(\mathbf{x}_i^{(r)}, \mathbf{x}_j^{(r)} \right)$$

Smaller Value Signifies a More Compact Cluster

Silhouette Measure: Inter-Cluster Distance

$D \left(\mathbf{x}_i^{(r)}, \mathbf{S}_l \right)$ is the Distance between $\mathbf{x}_i^{(r)}$ Cluster $\mathbf{S}_l$

$$D \left(\mathbf{x}_i^{(r)}, \mathbf{S}_l \right) = \frac{1}{n_l} \sum_{q=1}^{n_l} d \left(\mathbf{x}_i^{(r)}, \mathbf{x}_q^{(l)} \right)$$

Higher Value Implies that $\mathbf{S}_l$ is a Distant Cluster

Silhouette Measure: Inter-Cluster Distance

$b_i^{(r)}$ is the Distance between $\mathbf{x}_i^{(r)}$ and the Nearest Cluster

$$b_i^{(r)} = \min_{\substack{l=1 \dots K \\ l \neq r}} D \left(\mathbf{x}_i^{(r)}, \mathbf{S}_l \right) = \min_{\substack{l=1 \dots K \\ l \neq r}} \frac{1}{n_l} \sum_{q=1}^{n_l} d \left(\mathbf{x}_i^{(r)}, \mathbf{x}_q^{(l)} \right)$$

Higher Value Implies Significant Distance from Other Clusters

Silhouette Measure

$$s_i^{(r)} = \frac{b_i^{(r)} - a_i^{(r)}}{\max \{a_i^{(r)}, b_i^{(r)}\}} = \begin{cases} 1 - \frac{a_i^{(r)}}{b_i^{(r)}} & \text{IF } a_i^{(r)} < b_i^{(r)} \\ 0 & \text{IF } a_i^{(r)} = b_i^{(r)} \\ \frac{b_i^{(r)}}{a_i^{(r)}} - 1 & \text{IF } a_i^{(r)} > b_i^{(r)} \end{cases}$$

The First Case for $\bar{s}$ Signifies Good Clustering



Thank You